

SMARTTRACE: THE DIGITAL- RED THREAD OF FATE

AG 39 | AGGLOMERATION 2.0

CORE PROJECT OBJECTIVES

- Serialization & Label Generation Engine Investing & Trading
- Multilevel Hierarchy Management
- Real-Time Validation & Verification
- Anti-Counterfeit Protections
- User Interface

PILLAR 1 – LABEL GENERATION ENGINE

- **GS1 Compliance:** Generating SSCCs for tertiary packaging (Pallets)
- **Multi-Level Logic:** Automated prefixing—PRD1 (Primary), CRT (Secondary), and SSCC (Tertiary)
- **Data Integrity:** Implementation of a custom **Check Digit Algorithm** (Weighted Sum) to detect manual entry errors and tampering
- **Global Uniqueness:** Combining Company Prefix + Timestamp + Sequential Counter + Check Digit.

PILLAR 2 – HIERARCHY MANAGEMENT

- **Relationship Mapping:** Explicit tracking of Primary (Unit) -> Secondary (Carton) -> Tertiary (Pallet)
- **Configurable Aggregation:** Logic supports batch rules, such as 10 units per carton
- **Database Design:** Using an aggregation table to map parent-child IDs efficiently for rapid traversal

PILLAR 3 – VALIDATION & VERIFICATION

- **Verification API:** Dedicated endpoints for real-time check digit and identity validation.
- **Structural Verification:** Confirms if a specific unit truly belongs to the claimed secondary package
- **Fraud Prevention:** Built-in duplicate detection to flag if a serial number has been scanned before.

PILLAR 4 – ANTI-COUNTERFEIT FEATURES

- **Cryptographic Layer:** Generating SHA256 hashes using: Serial+ Date + Product code + Secret Salt
- **Verification Codes:** Storing the full hash in the DB while printing the first 8 characters on the label for quick verification.
- **Status Indicators:** Returns **Valid** (Pass), **Invalid** (Hash mismatch), or **Suspect** (Multiple scans)
- **Anomaly Detection:** Location-based tracking to identify counterfeit products scanned from different locations

PILLAR 5 – USER INTERFACE (UI/UX)

- **Generation Dashboard:** A professional interface to trigger batch generation and monitor progress in real-time
- **Analytics Dashboard:** Visual reporting of verification statistics, scan successes, and counterfeit hotspots
- **Verification:** Through User Input we Verify if the input is a suspicious input through Luhn and GS1 algorithms

SUCCESS CRITERIA & PROJECT MILESTONES

- **Performance:** Achieved generation of 10,000+ labels in < 5 seconds
- Guaranteed **Zero duplicate serial numbers** and 100% check digit accuracy
- **Security:** Successfully identifies tampered labels and logs security anomalies
- **UX:** Delivered a clean, demonstrable UI for both supply chain managers and field verifiers



THANK YOU

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